

- 2 (a) For a system in rotational equilibrium, sum of clockwise moments about any point equals sum of anticlockwise moments about the same point [1]
- (b) (i) Taking moment about the pivot, [1]
 $2.6 \times 0.4 = (4.0 - U) \times 0.4$ [1]
 $U = (4.0 - 2.6)$
 $= 1.4 \text{ N}$
- (ii) $U = \rho g V$ and $A = V / h$ [1]
or
 $p = h \rho g$ and $A = U / p$
or
 $A = U / h \rho g$
- $A = 1.4 / (0.10 \times 1.0 \times 10^3 \times 9.81)$
 $= 1.43 \times 10^{-3} \text{ m}^2$
 $= 1.4 \times 10^{-3} \text{ m}^2$ [1]
- (c) line starting at (0.10, 0.40) [1]
 straight line with negative gradient [1]
 line ending at (0.28 to 0.29, 0) [1]



